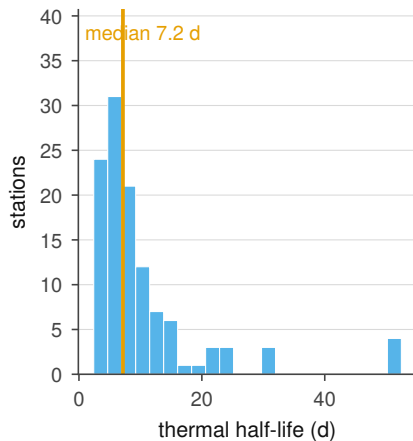
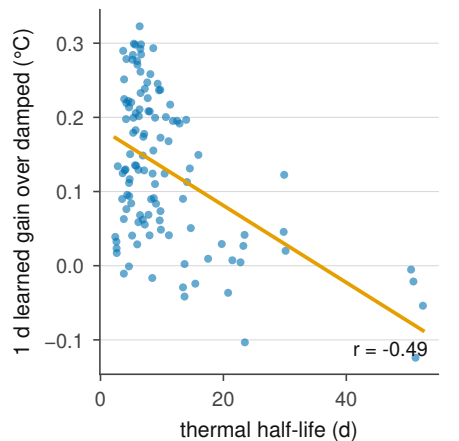


(a) Thermal memory

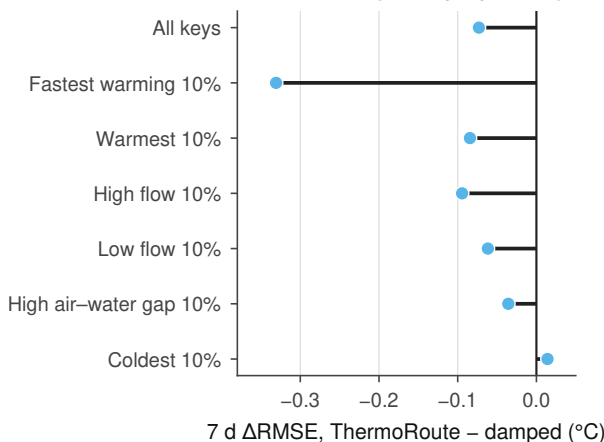


(b) Longer memory, less to learn

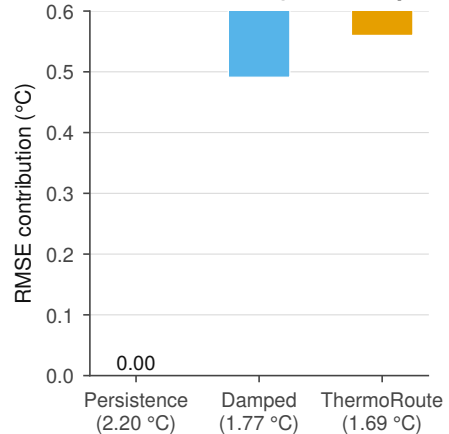


0.49

(c) Learned gain by hydrologic state



(d) 7 d error-budget decomposition



Half-life from a training-period AR(1) on damped-seasonal anomalies; learned gain is the station-median RMSE difference damped minus ThermoRoute; stratified differences are pooled over the held-out keys within each state (Section 4.8).